Systematic review of Tuberculosis prevalence in low- and middle-income countries

Extraction form

1. Study identification

1.1. Study DOI:

Please input only the unique DOI and not the full URL.
For example, "10.1371/journal.pgph.0000784".
If extracting data from multiple publications, input the primary prevalence survey DOI first, followed by the DOI for each additional paper, separated by a semi-colon. Only include those papers from which you have extracted data.
Write NA if not provided.

1.2. First Author's surname:

1.3. Paper Title:

1.4. Publication Year:

2. Corresponding Author

2.1. Corresponding author (surname, initials):

2.2. Corresponding author's email address:

Record phone number or mailing address if no email address is provided.

3. Study methodology

If both adults and children were included in the study population, answer the following questions regarding methods used for adult participants.

3.1. Does the study report that it has adhered to guidelines from the WHO handbook for prevalence surveys?

1. **Yes**
2. **No**
3. **Unknown**

3.2. Selection of study population

1. **Recruitment of all individuals within one or more areas**
2. **Simple random sampling within one cluster**
3. **Simple random sampling within >1 cluster**

4. **Multistage random sampling within >1 cluster (with or without weighting)**

3.2.1. Does this study include multiple prevalence surveys?

1. **No**
2. **Yes (If yes, you will be directed to another page to fill in requisite details)**
3. **Unsure (requires discussion with team)**

3.3. Country(ies) of study:
If multiple countries, separate each with a comma and a space

3.4. Study Setting:
Provide any contextual / geographical details of where the study was conducted.

3.5. What was the geographic coverage of the study?

1. **Single area (e.g., province, district, city)**
2. **Multiple areas (but not nationally representative)**
3. **Nationally representative**
4. **Other**

3.6 What was the urban/rural coverage of the study?

1. **Urban only**
2. **Rural only**
3. **Urban and rural**
4. **Unknown**

3.7. Study Year(s) (time-period during which study was conducted):
Record four-digit year; if multiple years then use format '9999-9999.'

3.8. Were contact investigations used to identify additional participants?
Make sure to exclude mass-surveys around circumference of homes.

1. **Yes**
2. **No**
3. **Unknown**

4. Screening

4.1. Screening criteria
Description (optional)

4.2. Were individuals currently on TB treatment included in the study population?

1. **Yes**
2. **No**
3. **Unknown**

4.3. Were participants screened prior to diagnosis?

1. **Yes**
2. **No**

4.4. Which of the following screening tests were done? (Select those that apply.)
Example for "Other": Latent TB test

1. **Chest X-ray only**
2. **Symptom screen only**
3. **Symptom screen, then chest X-ray (if positive)**
4. **Chest X-ray, then symptom screen (if positive)**
5. **Chest X-ray and symptom screen**
6. **No screening performed**
7. **Other**

4.5. Answer if symptom screen was performed. Symptom screen positive defined by: (Select those that apply.)

1. **Cough ≥ 3 weeks**
2. **Cough ≥ 2 weeks**
3. **Cough of other or unknown duration**
4. **Sputum production**
5. **Haemoptysis**
6. **Chest pain**
7. **Fever**
8. **Night sweats**
9. **Weight loss**
10. **Symptom screen was not performed**
11. **Other**

4.6. If chest X-ray was used, how was a positive result defined?

1. **Any abnormality**
2. **Indicative of TB**
3. **Chest X-ray was not used**
4. **Other**

5. Diagnostics

5.1. From which participants was sputum collected?

1. **All participants**
2. **Participants with symptoms**
3. **Participants with abnormal chest X-ray**
4. **Participants with symptoms OR abnormal chest X-ray**
5. **No sputum collected**
6. **Unknown**
7. **Other**

5.2. Were sputum samples tested by smear microscopy?

1. **Yes**
2. **No**
3. **No sputum collected**

4. **Unknown**

5.3. Which samples were tested by smear microscopy?

1. **All samples**
2. **Subsets of samples (Specify in 5.3.1)**
3. **No samples tested by smear microscopy**
4. **Unknown**
5. **Other**

5.3.1. If "Subsets of samples" was selected in 5.3, specify:

5.4. Which microscopy method was used?

1. **Light microscopy**
2. **Fluorescence**
3. **Light and Fluorescence microscopy**
4. **No samples tested by smear microscopy**
5. **Unknown**
6. **Other**

5.5. Were samples tested by Xpert MTB/RIF, or Xpert Ultra?

1. **Yes (Xpert MTB/RIF only)**
2. **Yes (Xpert Ultra only)**
3. **Yes (both Xpert MTB/RIF and Xpert Ultra, or either of these)**
4. **Neither**
5. **Unknown**

5.6. Which samples were tested by Xpert?

1. **All samples**
2. **Subset of samples (Specify in 5.6.1)**
3. **No samples tested by Xpert**
4. **Unknown**
5. **Other**

5.6.1. If "Subset of samples" was selected in 5.6, specify:

5.6.2. Were Xpert trace results classified as "Microbiologically-confirmed TB"?

1. **Yes**
2. **No**
3. **No samples tested by Xpert**

5.7. Were sputum samples tested by culture?

1. **Yes**
2. **No**
3. **Unknown**

5.8. Which samples were tested by culture?

1. **All samples**
2. **Subset of samples (Specify in 5.8.1)**
3. **No samples tested by culture**
4. **Unknown**
5. **Other**

5.8.1. If "Subset of samples" was selected in 5.8, specify:

5.9. Which culture media was used?
1. **Solid**
2. **Liquid**
3. **Solid and liquid**
4. **No samples tested by culture**
5. **Unknown**

5.9.1 If none of the above, which test was used?

6. Case Definitions

Give explanations for any case definitions used in the study. Write "N/A" for those that are not used in the study.

6.1. Presumptive TB (may be called "suspect TB" or "TB suspect" or "possible TB case", or similar in some reports)

6.2. Radiologically-confirmed TB

6.3. Bacteriologically-confirmed TB (may be called "microbiologically-confirmed TB", or similar in some reports)

6.4. Culture-positive TB

6.5. Sputum smear-positive TB

6.6. Sputum smear-negative TB

6.7. TB without bacteriological confirmation

6.8. Prevalent TB (please specify if treatment was provided or not)

6.9. If given, provide the definition of Rural vs. Urban participants

6.10. If given, provide the definition for how HIV status was ascertained.
1. Self-report
2. HIV test
3. Not specified
4. HIV status was not assessed

6.11. Any other case definition(s)

6.12. Additional comments (optional)

6.13. Were prevalence estimates adjusted in some way (synonyms: “standardized”, “weighted”, “controlled”)? If so, describe how. If not, enter, “no”.

6.14. Additional comments on study methodology

7. Results

If both adults and children were included in the study population, answer the following questions regarding adults only.

7.1. Participant description (optional)

7.2. What is the total target population of the survey area(s)?
Enter a number.

7.3. Does this study report results by sex?
If no, fill out the total column and leave other cells blank. Please write "N/A" if a cell in the "total" column
is unreported**.** If yes, please fill out table 7.3.1 completely, entering "N/A" if a value is not reported.

1. **Yes**
2. **No**

7.3.1 Results (by sex)
If a data cell is not reported, please type "N/A".

	Male	**Female**	**Total**
7.3.1. Eligible participants			
7.3.2. Participants			
7.3.3. People with presumptive TB			
7.3.4. People with TB symptoms			
7.3.5. People with abnormal chest X-ray			
7.3.6. Persons with successful sputum samples			
7.3.7. Radiologically-confirmed TB [bacteriological unconfirmed]			
7.3.8. Bacteriologically-confirmed TB			
7.3.9. Sputum smear-positive TB			
7.3.10. Culture-positive TB			

	Male	Female	Total
7.3.11. Prevalent TB			
7.3.12. [Crude] Prevalence of bacteriologically-confirmed TB (per 100,000)			
7.3.13. [Crude] Confidence Interval of 7.3.12			
7.3.14. [Crude] Prevalence of sputum smear-positive TB (per 100,000)			
7.3.15. [Crude] Confidence Interval of 7.3.14			
7.3.16. [Crude] Prevalence of all forms of TB (per 100,000)			
7.3.17. [Crude] Confidence Interval of 7.3.16			
7.3.18. [Adjusted] Prevalence of bacteriologically-confirmed TB (per 100,000)			
7.3.19. [Adjusted] Confidence Interval of 7.3.18			
7.3.20 [Adjusted] Prevalence of sputum smear-positive TB (per 100,000)			
7.3.21 [Adjusted] Confidence Interval of 7.3.20			
7.3.22 [Adjusted] Prevalence of all forms of TB (per 100,000)			
7.3.23 [Adjusted] Confidence Interval of 7.3.22			

7.4 Does this study report results by urban/rural stratification?

If no, leave all cells blank and move to 7.5.

If yes, please fill out table 7.4.1 completely, entering "N/A" if a value is not reported.

1. **Yes**
2. **No**

7.4.1 Results (by rurality)

If a data cell is not reported, please type "N/A".

	Urban	Rural	Total
7.4.1. Eligible participants			
7.4.2. Actual participants			
7.4.3. People with presumptive TB			
7.4.4. People with TB symptoms			
7.4.5. People with abnormal chest X-ray			
7.4.6. Persons with successful sputum samples			

	Urban	Rural	Total
7.4.7. Radiologically-confirmed TB [bacteriological unconfirmed]			
7.4.8. Bacteriologically-confirmed TB			
7.4.9. Sputum smear-positive TB			
7.4.10. Culture-positive TB			
7.4.11. Prevalent TB			
7.4.12. [Crude] Prevalence of bacteriologically-confirmed TB (per 100,000)			
7.4.13. [Crude] Confidence Interval of 7.4.12			
7.4.14. [Crude] Prevalence of sputum smear-positive TB (per 100,000)			
7.4.15. [Crude] Confidence Interval of 7.4.14			
7.4.16. [Crude] Prevalence of all forms of TB (per 100,000)			
7.4.17. [Crude] Confidence Interval of 7.4.16			
7.4.18. [Adjusted] Prevalence of bacteriologically-confirmed TB (per 100,000)			
7.4.19. [Adjusted] Confidence Interval of 7.4.18			
7.4.20. [Adjusted] Prevalence of sputum smear-positive TB (per 100,000)			
7.4.21. [Adjusted] Confidence Interval of 7.4.20			
7.4.22 [Adjusted] Prevalence of all forms of TB (per 100,000)			
7.4.23. [Adjusted] Confidence Interval of 7.4.22			

7.5 Does this survey/study report results by HIV status?

If no, leave all cells blank and move to 7.6.
If yes, please fill out table 7.5.1 completely, entering "N/A" if a value is not reported.

1. Yes
2. No

7.5.1 Results (by HIV status)

If a data cell is not reported, please type "N/A".

	HIV-positive	HIV-negative	Total
7.5.1. Eligible participants			
7.5.2. Participants			

	HIV-positive	HIV-negative	Total
7.5.3. People with presumptive TB			
7.5.4. People with TB symptoms			
7.5.5. People with abnormal chest X-ray			
7.5.6. Persons with successful sputum samples			
7.5.7. Radiologically-confirmed TB [bacteriological unconfirmed]			
7.5.8. Bacteriologically-confirmed TB			
7.5.9. Sputum smear-positive TB			
7.5.10. Culture-positive TB			
7.5.11. Prevalent TB			
7.5.12. [Crude] Prevalence of bacteriologically-confirmed TB (per 100,000)			
7.5.13. [Crude] Confidence Interval of 7.5.12			
7.5.14. [Crude] Prevalence of sputum smear-positive TB (per 100,000)			
7.5.15. [Crude] Confidence Interval of 7.5.14			
7.5.16. [Crude] Prevalence of all forms of TB (per 100,000)			
7.5.17. [Crude] Confidence Interval of 7.5.16			
7.5.18. [Adjusted] Prevalence of bacteriologically-confirmed TB (per 100,000)			
7.5.19. [Adjusted] Confidence Interval of 7.5.18			
7.5.20. [Adjusted] Prevalence of sputum smear-positive TB (per 100,000)			
7.5.21. [Adjusted] Confidence Interval of 7.5.20			
7.5.22 [Adjusted] Prevalence of all forms of TB (per 100,000)			
7.5.23. [Adjusted] Confidence Interval of 7.5.22			

7.6 Does this survey/study report results by age group?

If no, leave all cells blank and move to 7.7.

If yes, please fill out table 7.6.1 completely, entering "N/A" if a value is not reported.

1. Yes
2. No

7.6.1 Results (by age group)

If a data cell is not reported, please type "N/A".

	0-14	15-24	25-34	35-44	45-54	55-64	65+	Total
7.6.1. Age-group definitions, if different (format "XX-YY")								
7.6.2. Eligible participants								
7.6.3. Participants								
7.6.4. People with presumptive TB								
7.6.5. People with TB symptoms								
7.6.6. People with abnormal chest X-ray								
7.6.7. Persons with successful sputum samples								
7.6.8. Radiologically-confirmed TB [bacteriological unconfirmed]								
7.6.9. Bacteriologically-confirmed TB								
7.6.10. Sputum smear-positive TB								
7.6.11. Culture-positive TB								
7.6.12. Prevalent TB								
7.6.13. [Crude] Prevalence of bacteriologically-confirmed TB (per 100,000)								
7.6.14. [Crude] Confidence Interval of 7.6.13								
7.6.15. [Crude] Prevalence of sputum smear-positive TB (per 100,000)								
7.6.16. [Crude] Confidence Interval of 7.6.15								
7.6.17. [Crude] Prevalence of all forms of TB (per 100,000)								
7.6.18. [Crude] Confidence Interval of 7.6.17								
7.6.19. [Adjusted] Prevalence of bacteriologically-confirmed TB (per 100,000)								
7.6.20. [Adjusted] Confidence Interval of 7.6.19								
7.6.21. [Adjusted] Prevalence of sputum smear-positive TB (per 100,000)								
7.6.22. [Adjusted] Confidence Interval of 7.6.21								

	0-14	15-24	25-34	35-44	45-54	55-64	65+	Total
7.6.23 [Adjusted] Prevalence of all forms of TB (per 100,000)								
7.6.24. [Adjusted] Confidence Interval of 7.6.23								

7.7 Does this study report results by both sex and urban/rural stratification?
If no, leave all cells blank and move to 7.8.
If yes, please fill out table 7.7.1 completely, entering "N/A" if a value is not reported.

1. Yes
2. No

7.7.1 Results (by sex and rurality)

If a data cell is not reported, please type "N/A".

	Male - Urban	Female - Urban	Male - Rural	Female - Rural	Total
7.7.1. Eligible participants					
7.7.2. Participants					
7.7.3. People with presumptive TB					
7.7.4. People with TB symptoms					
7.7.5. People with abnormal chest X-ray					
7.7.6. Persons with successful sputum samples					
7.7.7. Radiologically-confirmed TB [bacteriological unconfirmed]					
7.7.8. Bacteriologically-confirmed TB					
7.7.9. Sputum smear-positive TB					
7.7.10. Culture-positive TB					
7.7.11. Prevalent TB					
7.7.12. [Crude] Prevalence of bacteriologically-confirmed TB (per 100,000)					
7.7.13. [Crude] Confidence Interval of 7.7.12					
7.7.14. [Crude] Prevalence of sputum smear-positive TB (per 100,000)					
7.7.15. [Crude] Confidence Interval of 7.7.14					
7.7.16. [Crude] Prevalence of all forms of TB (per 100,000)					

	Male - Urban	Female - Urban	Male - Rural	Female - Rural	Total
7.7.17. [Crude] Confidence Interval of 7.7.16					
7.7.18. [Adjusted] Prevalence of bacteriologically-confirmed TB (per 100,000)					
7.7.19. [Adjusted] Confidence Interval of 7.7.18					
7.7.20. [Adjusted] Prevalence of sputum smear-positive TB (per 100,000)					
7.7.21. [Adjusted] Confidence Interval of 7.7.20					
7.7.22 [Adjusted] Prevalence of all forms of TB (per 100,000)					
7.7.23. [Adjusted] Confidence Interval of 7.7.22					

7.8. Additional comments on results and/or data availability.

7.9. List any additional references that should be examined for inclusion in this systematic review.
Copy the full citation from the study's reference list; skip a line between each reference.

8. Study Quality

See Hoy et al. (2012) Appendix for examples and additional details on study quality assessment.

8.1. Was the study's sample population a true or close representation of the target population in relation to relevant variables, e.g., age, sex, occupation?

1. **Yes (low risk)**
2. **No (high risk)**
3. **Unknown**

8.2. Was some form of random selection used to select the sample or was a census undertaken?

1. **Yes (low risk)**
2. **No (high risk)**

8.3. Was the likelihood of non-response bias minimal?

1. **Yes (low risk)**
2. **No (high risk)**

8.4. Were data collected directly from subjects (as opposed to a proxy)?

1. **Yes (low risk)**
2. **No (high risk)**

8.5. Was an acceptable case definition used in the study?

1. **Yes (low risk)**
2. **No (high risk)**

8.6. Was the study instrument that measured the parameter of interest shown to have reliability and validity (i.e., diagnostic methods)?

1. **Yes (low risk)**
2. **No (high risk)**

8.7. Was the same mode of data collection used for all subjects?

1. **Yes (low risk)**
2. **No (high risk)**

8.8. Were the numerator and denominator for the parameter of interest appropriate?

1. **Yes (low risk)**
2. **No (high risk)**

8.9. Summary item on the overall risk of bias (based on 8.1-8.8)

1. **Low risk of bias - Further research is very unlikely to change our confidence in the estimate**
2. **Moderate risk of bias - Further research is likely to have an important impact on our confidence in the estimates**
3. **High risk of bias - Further research is very likely to have an important impact on our confidence in the estimates**

8.10. Additional comments on study quality